Timing a possible encounter

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Learning target: Separate orbital period from an encounter time.

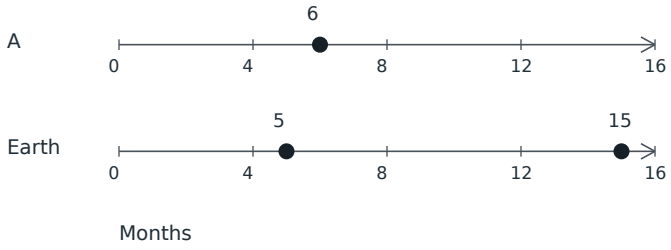
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 12.*
No. Tangent velocity arrows change direction, so inward acceleration and net force are needed.

02. Worked example: explain the model
Month 14 matches; a simultaneous encounter requires nearby positions at the same time.

03. Read and interpret the model
No; the given times differ.

Model / diagram check



04. Apply the physics
Years 8 and 16; no Earth location not given.

05. Explain your reasoning
A bound is not a probability distribution; need joint timing/position uncertainty and encounter geometry.

06. Review: connect an earlier idea *Review: Unit 2, Day 3.*
Mass 4 kg; weight 24 N toward the source. Weight depends on field.

07. Exit check
Orbital phase/time reference and both positions/geometry; uncertainty also valid.

08. STRETCH: test the model
Example A at months 1, 5, 9 and B at 3, 7, 11; no coincidence in stated interval.

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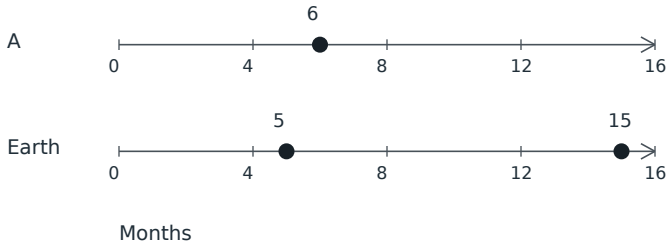
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 12.*
Both become 1/3; $a_c = v^2/r$ and $F_{net} = mv^2/r$.

02. Worked example: explain the model
Month 14 matches; a simultaneous encounter requires nearby positions at the same time.

03. Read and interpret the model
No; the given times differ.

Model / diagram check



04. Apply the physics
6 and 42 months. Ideal periods/geometry fixed, does not include uncertainty, 3 D miss distance, etc.

05. Explain your reasoning
A bound is not a probability distribution; need joint timing/position uncertainty and encounter geometry.

06. Review: connect an earlier idea *Review: Unit 2, Day 3.*
Mass 4 kg; new weight 24 N.

07. Exit check
Days 13-14 overlap; possible timing consistency, not proof of collision without spatial geometry/uncertainty.

08. STRETCH: test the model
First common time 19 months (n=3, m=2). Ignores miss distance, changing periods and uncertainty; timing alone does not prove impact.